

## Mass Conservation

Applying mass conservation on an elemental control volume for a constituent  $i$ :

$$\underbrace{\frac{\partial \rho_i}{\partial t}}_{\text{Accumulation}} + \underbrace{\frac{\partial}{\partial x} (\rho_i u_i) + \frac{\partial}{\partial y} (\rho_i v_i)}_{\text{Net mass flux}} = \underbrace{m_i'''}_{\text{Mass Generation } \left[\frac{\text{kg}}{\text{m}^3 \cdot \text{s}}\right]}$$

Note, we get all of our old results back if summing over all constituents and let  $m_i''' = 0$ .

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} \sum \rho_i u_i + \frac{\partial}{\partial y} \sum \rho_i v_i = 0}$$

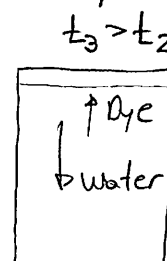
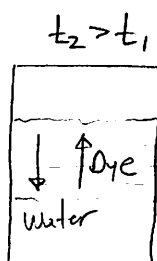
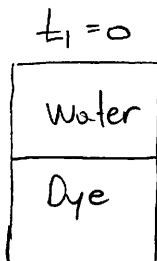
$\Downarrow$  Has to be true for an equilibrium mixture

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) = 0} \Rightarrow \text{Derived in 1st part of class.}$$

So from this, we can say:

$$\boxed{u = \frac{1}{\rho} \sum \rho_i u_i} ; \boxed{v = \frac{1}{\rho} \sum \rho_i v_i} \equiv \text{Mass averaged velocity}$$

Note,  $u \neq u_i$  or  $v \neq v_i$ . Think of dye diffusing in water



$$\begin{aligned} V_{\text{dye}} &= V_1 > 0 \\ V_{\text{water}} &= V_2 < 0 \\ V &= 0 \end{aligned}$$

While the mass averaged velocity ( $V$ ) is zero, the velocity of the constituents is not.

$(u_i - u) \equiv$  Diffusion Velocity in x-direction  
 $\rho_i(u_i - u) \equiv$  Flow rate per unit area of  $i$  in x-direction relative to the bulk motion of the mixture.

$$\boxed{\begin{aligned} j_{x,i} &= \rho_i(u_i - u) \\ j_{y,i} &= \rho_i(v_i - v) \end{aligned}} \equiv \text{Diffusive Flux}$$

Back substituting into our initial constituent mass conservation

$$\frac{\partial \rho_i}{\partial t} + \frac{\partial}{\partial x}(\rho_i u) + \frac{\partial}{\partial y}(\rho_i v) = -\frac{\partial j_{x,i}}{\partial x} - \frac{\partial j_{y,i}}{\partial y} + m_i'''$$

Noting that  $C_i = \rho_i$  and  $\rho = \text{constant}$  (incompressible)

$$\frac{\partial C_i}{\partial t} + u \frac{\partial C_i}{\partial x} + v \frac{\partial C_i}{\partial y} = -\frac{\partial j_{x,i}}{\partial x} - \frac{\partial j_{y,i}}{\partial y} + m_i'''$$

or

$$\boxed{\frac{DC_i}{Dt} = -\nabla \cdot \mathbf{j}_i + m_i'''} \Rightarrow \text{Looks Familiar?}$$

The diffusion flux vector  $\mathbf{j}_i$  is driven by  $\nabla C_i$ , same as the heat flux vector  $\mathbf{q}''$  is driven by  $\nabla T$ .

In 1855, Fick noticed this and mass transfer was born:  
 For a 2 component mixture: (dilute approximation,  $\rho_{\text{tot}} = \text{const}$ )

$$\boxed{\mathbf{j}_1 = -D_{12} \nabla C_1} \quad ; \quad D_{12} = D_{21} = \overset{\substack{\text{[m}^2/\text{s}]}{\downarrow}}{D} \equiv \text{Mass Diffusivity of component 1 into component 2.}$$

Back substituting  $\mathbf{j}_i$  into mass conservation above:

$$\boxed{\frac{DC}{Dt} = D \nabla^2 C + \dot{m}'''} \equiv \text{Dropped the subscript } i$$

$\hookrightarrow$  Just like heat transfer

Note, in heat transfer we had:

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{\dot{q}'''}{\rho c_p} \equiv \text{Energy Equation}$$

$$\dot{q}'' = -k \nabla T$$

$$\boxed{C \Rightarrow T}, \quad \boxed{D \Rightarrow \alpha}$$

So we can solve everything through analogy.  
For cartesian

$$\boxed{\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} + w \frac{\partial C}{\partial z} = D \left( \frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2} \right) + \dot{m}'''} \quad \text{}$$

For mass fraction:

$$\boxed{\frac{\partial \phi}{\partial t} = D \nabla^2 \phi + \frac{\dot{m}'''}{\rho}}$$

$$\boxed{j = -\rho D \nabla \phi}$$

For mole fraction:

$$\boxed{\frac{\partial x}{\partial t} = D \nabla^2 x + \frac{M}{M_i} \frac{\dot{m}'''}{\rho}}$$

$$\boxed{j = -\rho \frac{M_i}{M} D \nabla x}$$

$\Rightarrow M_i$  is molar mass of constituent of interest, whose mole fraction is  $x$ .

Mass Diffusivities (Binary Mixtures)

See Table 11.1 of pg. 498 in Bejan for  $D_{12} = D_{21} = D$  at  $T_0, P_0$

$$\boxed{\frac{D(T, P)}{D(T_0, P_0)} \approx \left( \frac{T}{T_0} \right)^{1.75} \frac{P_0}{P}}$$

\* Note,  $D$  for liquid mixtures is  $10^4$  to  $10^5$  smaller than for gases

**Table 11.1 Mass diffusivities of binary gaseous mixture  
atmospheric pressure**

| Gaseous Mixture            | $D \text{ (m}^2\text{/s)}$ | $T \text{ (K)}$ |
|----------------------------|----------------------------|-----------------|
| Air-acetone                | $1.09 \times 10^{-5}$      | 273             |
| Air-ammonia                | $2.80 \times 10^{-5}$      | 298             |
| Air-benzene                | $0.77 \times 10^{-5}$      | 273             |
| Air-carbon dioxide         | $1.42 \times 10^{-5}$      | 276             |
|                            | $1.77 \times 10^{-5}$      | 317             |
| Air-ethanol                | $1.45 \times 10^{-5}$      | 313             |
| Air-helium                 | $7.65 \times 10^{-5}$      | 317             |
| Air- <i>n</i> -hexane      | $0.80 \times 10^{-5}$      | 294             |
| Air-methanol               | $1.32 \times 10^{-5}$      | 273             |
| Air-naphthalene            | $5.13 \times 10^{-6}$      | 273             |
| Air-water vapor            | $2.60 \times 10^{-5}$      | 298             |
|                            | $2.88 \times 10^{-5}$      | 313             |
| Ammonia-hydrogen           | $5.70 \times 10^{-5}$      | 263             |
|                            | $1.10 \times 10^{-4}$      | 358             |
| Argon-carbon dioxide       | $1.33 \times 10^{-5}$      | 276             |
| Argon-hydrogen             | $8.29 \times 10^{-5}$      | 295             |
| Benzene-hydrogen           | $4.04 \times 10^{-5}$      | 311             |
| Benzene-nitrogen           | $1.02 \times 10^{-5}$      | 311             |
| Carbon dioxide-nitrogen    | $1.67 \times 10^{-5}$      | 298             |
| Carbon dioxide-oxygen      | $1.53 \times 10^{-5}$      | 293             |
| Carbon dioxide-water vapor | $1.98 \times 10^{-5}$      | 307             |
| Cyclohexane-nitrogen       | $0.73 \times 10^{-5}$      | 288             |
| Helium-methane             | $6.76 \times 10^{-5}$      | 298             |
| Hydrogen-nitrogen          | $7.84 \times 10^{-5}$      | 298             |
| Hydrogen-water vapor       | $9.15 \times 10^{-5}$      | 307             |
| Methane-water vapor        | $3.56 \times 10^{-5}$      | 352             |
| Nitrogen-water vapor       | $3.59 \times 10^{-5}$      | 352             |
| Oxygen-water vapor         | $3.52 \times 10^{-5}$      | 352             |

\*Table adapted from Bejan, *Convection Heat Transfer*, 4<sup>th</sup> Edition, pp. 498

**Table 11.2 Mass diffusivities of gases and organic concentrations in water (dilute aqueous solutions)**

| Solute          | Solvent | $D \text{ (m}^2\text{/s)}$ | $T \text{ (K)}$ |
|-----------------|---------|----------------------------|-----------------|
| Acetone         | Water   | $1.16 \times 10^{-9}$      | 293             |
| Air             | Water   | $2.5 \times 10^{-9}$       | 293             |
| Aniline         | Water   | $0.92 \times 10^{-9}$      | 293             |
| Benzene         | Water   | $1.02 \times 10^{-9}$      | 293             |
| Carbon dioxide  | Water   | $1.92 \times 10^{-9}$      | 298             |
| Chlorine        | Water   | $1.25 \times 10^{-9}$      | 298             |
| Ethanol         | Water   | $0.84 \times 10^{-9}$      | 298             |
| Ethylene glycol | Water   | $1.04 \times 10^{-9}$      | 293             |
| Glycerol        | Water   | $0.72 \times 10^{-9}$      | 288             |
| Hydrogen        | Water   | $4.5 \times 10^{-9}$       | 298             |
| Nitrogen        | Water   | $2.6 \times 10^{-9}$       | 293             |
| Oxygen          | Water   | $2.1 \times 10^{-9}$       | 298             |
| Propane         | Water   | $0.97 \times 10^{-9}$      | 293             |
| Urea            | Water   | $1.2 \times 10^{-9}$       | 293             |
| Vinyl chloride  | Water   | $1.34 \times 10^{-9}$      | 298             |

\*Table adapted from Bejan, Convection Heat Transfer, 4<sup>th</sup> Edition, pp. 499

**Table 11.3 Henry's constant  $H$  for several gases in water at moderate pressures**

| $T \text{ (K)}$ | $H \text{ (bar)}$  |                    |                   |                   |                   |                   |
|-----------------|--------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
|                 | Air                | $N_2$              | $O_2$             | $H_2$             | $CO_2$            | CO                |
| 290             | $6.2 \times 10^4$  | $7.6 \times 10^4$  | $3.8 \times 10^4$ | $6.7 \times 10^4$ | $1.3 \times 10^3$ | $5.1 \times 10^4$ |
| 300             | $7.4 \times 10^4$  | $8.9 \times 10^4$  | $4.5 \times 10^4$ | $7.2 \times 10^4$ | $1.7 \times 10^3$ | $6 \times 10^4$   |
| 320             | $9.2 \times 10^4$  | $1.1 \times 10^5$  | $5.7 \times 10^4$ | $7.6 \times 10^4$ | $2.7 \times 10^3$ | $7.4 \times 10^4$ |
| 340             | $1.04 \times 10^5$ | $1.24 \times 10^5$ | $6.5 \times 10^4$ | $7.6 \times 10^4$ | $3.7 \times 10^3$ | $8.4 \times 10^4$ |

\*Table adapted from Bejan, Convection Heat Transfer, 4<sup>th</sup> Edition, pp. 501

## Boundary Conditions

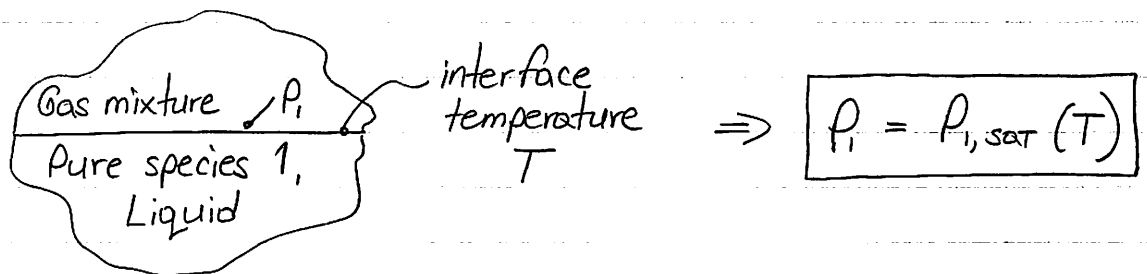
For mass transfer problems, B.C.'s are always applied on the inner side of boundaries facing the domain of mass transfer.

↗ concentration

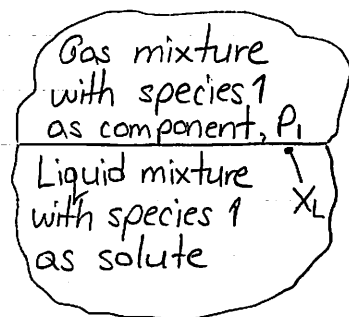
Unlike  $T$ ,  $C$  does not vary continuously across an interface

We can have 3 separate cases to consider:

Case 1: Ideal gas mixture and liquid phase of one component



Case 2: Interface between a liquid mixture and gas mixture



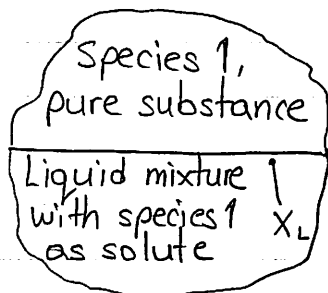
$X_L \equiv$  mole fraction of species 1 on liquid side  
 $P_i \equiv$  partial pressure of 1.  
 $\Rightarrow$  Henry's Law  
 $H \equiv$  Henry's constant

$$X_L = \frac{P_i}{H}$$

Look up  $H$  in Table 11.3 on pg. (188) of notes  
Valid for dilute solutions only.

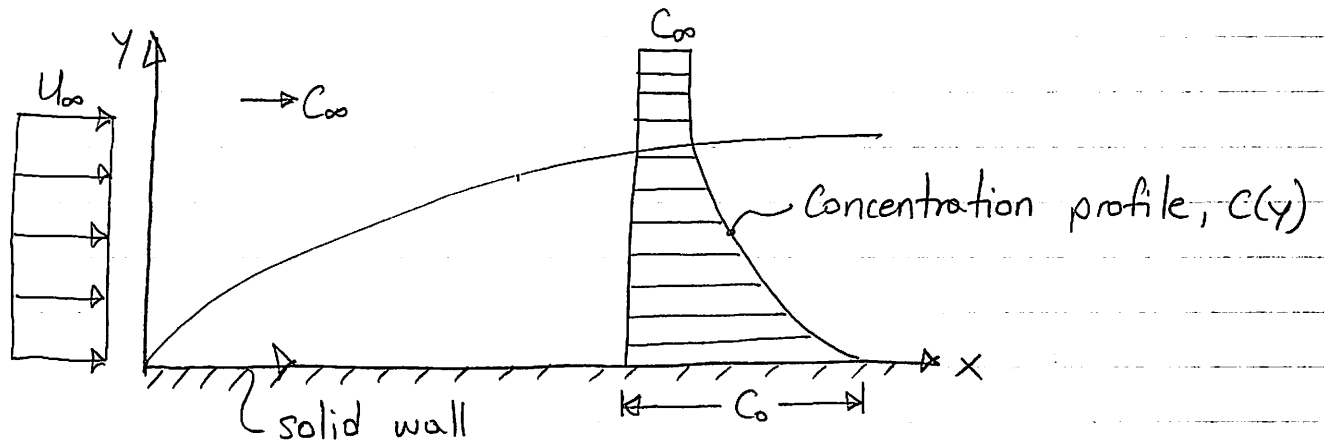
Henry's law also only valid at low pressures ( $P_i < 1 \text{ atm}$ ). At higher pressures,  $H = f(P_i)$ .

Cases: Interface between a liquid mixture and a pure species 1 (liquid, solid)



$\Rightarrow X_L$  determined by assuming thermodynamic equilibrium at the interface and looking up the solubility of species 1 in the solvent.

## Laminar Forced Convection Mass Transfer



Assuming we have a wet wall with humid air at  $C_\infty$  flowing past it. Assuming a slender b.l. with  $\frac{\partial p}{\partial x} = 0$  and  $T = \text{uniform everywhere}$ .

$$j_0 = -D \left( \frac{\partial C}{\partial y} \right) \bigg|_{y=0}$$

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \frac{\partial^2 C}{\partial y^2} \quad (1)$$

$$\text{B.C.'s: } \left. \begin{array}{l} C = C_0 \text{ at } y=0 \\ C \rightarrow C_\infty \text{ at } y \rightarrow \infty \end{array} \right\} (2)$$

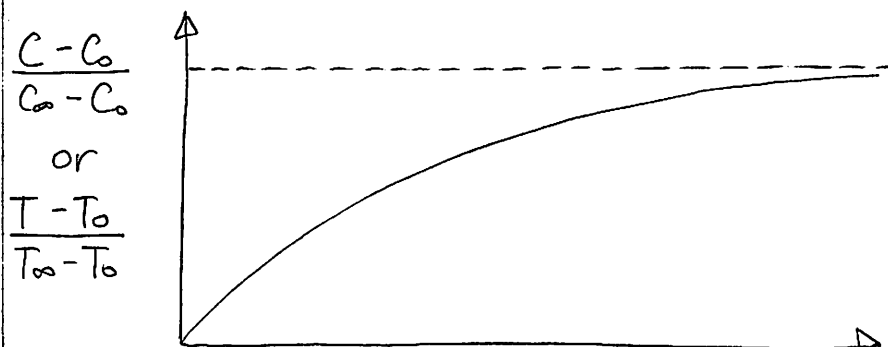
We know  $(u, v)$  from the Blasius solution (flow over a flat plate). The mass transfer problem (eq. ① & ②) is analogous to the Pohlhausen solution for heat transfer.

Noting that:  $T \rightarrow C$  and  $\alpha \rightarrow D$ :

$$\left[ \frac{\partial C}{\partial y} \right]_{y=0} = (C_\infty - C_0) \left( \frac{U_\infty}{\nu x} \right)^{1/2} \left\{ \int_0^\infty \exp \left[ -\frac{\nu D}{2} \int_0^\infty f(\beta) d\beta \right] d\eta \right\}^{-1}$$

↳ See page (55) of notes to see how we got this.

Graphically, we can draw this.



For the mass transfer solution:

$$\eta_m^* = \eta Sc^{1/3} \quad \text{where} \quad \boxed{Sc = \frac{U}{D}} = \text{Schmidt \#}$$

$U = \text{mixture viscosity}$

For the thermal solution, we had  $\eta^* = \eta Pr^{1/3}$ , so  $Sc \leftrightarrow Pr$ .  
Everything else is analogous!

$$\boxed{h_m = \frac{\dot{J}_0}{C_0 - C_\infty}} = \text{mass transfer coefficient}$$

All we need to do is replace  $Pr$  with  $Sc$ , &  $h$  with  $h_m$ :

$$\left. \begin{aligned} Nu &= \frac{hx}{k} = 0.332 Re_x^{1/2} Pr^{1/3} \\ \overline{Nu}_L &= \frac{\bar{h}L}{k} = 0.664 Re_L^{1/2} Pr^{1/3} \end{aligned} \right\} \text{Heat Transfer}$$

↓ Mass Transfer

$$\left. \begin{aligned} Sh &= \frac{h_m x}{D} = 0.332 Re_x^{1/2} Sc^{1/3} \\ \overline{Sh}_L &= \frac{\bar{h}_m L}{D} = 0.664 Re_L^{1/2} Sc^{1/3} \end{aligned} \right\} \text{Mass Transfer} \quad Sc > 0.5$$

$$\boxed{Sh \equiv \text{Sherwood \# (Analogous to Nusselt \#)}}$$

$$Nu \rightarrow Sh; \quad Pr \rightarrow Sc; \quad Re_x \rightarrow Re_x; \quad h \rightarrow h_m,$$

Lastly, we want the mass transfer rate,  $\dot{m}'$  (kg of species per second per meter depth):

$$\boxed{\dot{m}' = \bar{h}_m L (C_0 - C_\infty)} \quad ; \quad \boxed{\bar{h}_m = \frac{1}{L} \int_0^L h_m dx}$$

$$\boxed{\dot{m} = \bar{h}_m A (C_0 - C_\infty)}$$

### Impermeable Surface Model

Note, our previous solution relies on the Blasius model where we assumed  $v|_{y=0} = 0$ . For mass transfer problems, this may not be the case.

Let's try some scaling arguments:

$$j_0 \sim (C_0 - C_\infty) \frac{D}{x} Re_x^{1/2} Sc^n \quad ; \quad \begin{aligned} n &= \frac{1}{3} \text{ if } Sc > 0.5 \\ n &= \frac{1}{2} \text{ if } Sc < 0.5 \end{aligned}$$

So the velocity scale  $v$  associated with the mass flux is:

$$v_0 \sim \frac{j_0}{\rho} \quad (1)$$

For our laminar boundary layer, the vertical (transverse) velocity leaving the boundary layer is:

$$v_\infty \sim U_\infty Re_x^{-1/2} \quad (\text{see page (48) of notes}) \quad \begin{aligned} \frac{v}{U_\infty} &= \frac{1}{2} Re_x^{-1/2} (\eta f' - f) \\ \eta = \frac{y}{\delta(x)} &= 1 \text{ (at boundary)} \\ f' = \frac{v}{U_\infty} &= 1 \text{ (at boundary)} \\ f &\sim \frac{\eta^2}{2} \end{aligned}$$

We can say that the solution is valid if  $v_0 < v_\infty$

Back substituting:

$$\begin{aligned} \frac{v_0}{v_\infty} &\sim \frac{(C_0 - C_\infty) \frac{D}{x} Re_x^{1/2} Sc^n}{U_\infty Re_x^{-1/2}} \sim \left( \frac{C_0 - C_\infty}{\rho} \right) \frac{D}{U_\infty x} \cdot Re_x \cdot \left( \frac{v}{U_\infty} \right)^n \\ &\sim \left( \frac{C_0 - C_\infty}{\rho} \right) \frac{D^{1/2}}{U_\infty^{1/2}} \cdot \frac{U}{U_\infty x} \cdot Re_x \cdot Re_x^{-1} \sim \left( \frac{C_0 - C_\infty}{\rho} \right) \cdot Sc^{-1/2} < 1 \end{aligned}$$

$$\boxed{\left( \frac{C_0 - C_\infty}{\rho} \right) < Sc^{1/2} < 1} \Rightarrow \text{Valid for small concentration differences.}$$